SNR of a Multimode Mechanical Detector: Noise Transformation and the SNR Formula

Companion note

1 Notation guide

Every symbol has a single fixed meaning throughout. Derived quantities carry a reference to the equation where they are defined.

Symbol	Meaning	Defined
N	Number of oscillators (= number of modes)	—
m_i	Mass of oscillator i	—
f_n	Normal mode frequency [Hz]	eq. (1)
V_{in}	(i, n) entry of modal matrix $\mathbf{V}$: displacement of oscillator i in mode n	eq. (1)
μ_n	Modal mass [kg]	eq. (3)
γ_i	Viscous damping rate of oscillator i [s^{-1}]	eq. (7)
Γ_n	Modal linewidth [Hz]	eq. (7)
$\chi_n(f)$	Modal susceptibility [$\mathrm{m\,N}^{-1}$]	eq. (11)
$Q_n(t)$	Generalized force on mode n [N]	eq. (9)
w_i	Readout weight for oscillator i	eq. (12)
N_n	Modal readout coupling	eq. (14)
β_i	Signal–oscillator coupling [N]	eq. (19)
$\mathcal{B}_n$	Modal signal coupling [N]	eq. (21)
$T(f)$	Transfer function from h to O [m]	eq. (23)
$S_O^{\mathrm{noise}}(f)$	Total noise PSD at observable [$\mathrm{m^2\,Hz^{-1}}$]	eq. (18)
$S_h(f)$	Strain-referred noise PSD [Hz^{-1}]	eq. (17)
ϕ	Optical phase [rad]	eq. (30)
S_ϕ	Shot noise PSD on phase [$\mathrm{rad^2\,Hz^{-1}}$]	eq. (33)
η_{det}	Detector quantum efficiency	eq. (33)
$\mathcal{A}$	Inspiral strain amplitude [$\mathrm{strain\,Hz}^{7/6}$]	eq. (44)
$\mathcal{M}_c$	Chirp mass [kg]	eq. (44)
r	Luminosity distance [m]	eq. (44)
ι	Inclination angle [rad]	eq. (44)

2 Physical geometry

We consider N optically levitated particles arranged along the axis of an optical cavity. The cavity axis and the array axis both lie along $\hat{x}$. A gravitational wave drives motion of the particles along $\hat{x}$. The cavity probe beam propagates along $\hat{x}$ and is used to read out the x-displacements.

3 Mechanical setup

3.1 Modal decomposition

The normal mode frequencies f_n and mode shapes V_{in} are solutions of the generalized eigenvalue problem:

$$\mathbf{K}\,\mathbf{v}_n = (2\pi f_n)^2\,\mathbf{M}\,\mathbf{v}_n, \qquad n = 1, \ldots, N, \tag{1}$$

where $\mathbf{M} = \mathrm{diag}(m_i)$ and $\mathbf{K}$ is the stiffness matrix. The eigenvectors are real (since $\mathbf{K}$ is symmetric) and V_{in} denotes the i-th component of $\mathbf{v}_n$. The displacement of oscillator i is

$$x_i(t) = \sum_{n=1}^{N} V_{in}\,q_n(t), \tag{2}$$

where $q_n(t)$ is the modal coordinate. The modal mass is

$$\mu_n \equiv \mathbf{v}_n^T \mathbf{M}\,\mathbf{v}_n = \sum_{i=1}^{N} m_i\,V_{in}^2. \tag{3}$$

Mass and stiffness orthogonality,

$$\sum_{i,j} V_{im}\,M_{ij}\,V_{jn} = \mu_n\,\delta_{mn}, \tag{4}$$

$$\sum_{i,j} V_{im}\,K_{ij}\,V_{jn} = \mu_n\,(2\pi f_n)^2\,\delta_{mn}, \tag{5}$$

hold when $\mathbf{K}$ is symmetric and $\mathbf{M}$ is positive definite, and are what allow the equations of motion to be decoupled into independent scalar equations for each q_n. Non-symmetric $\mathbf{K}$ (arising from feedback couplings, optomechanical back-action, or gyroscopic effects) breaks both relations and requires a first-order $2N$-dimensional state-space treatment.

3.2 Damping and modal linewidth

We take the damping matrix $\mathbf{C}$ to be diagonal with entries $C_{ii} = m_i\gamma_i$, where γ_i is the viscous damping rate of oscillator i. The modal damping matrix has elements

$$(\mathbf{V}^T\mathbf{C}\mathbf{V})_{mn} = \sum_{i} V_{in}\,m_i\,\gamma_i\,V_{im}. \tag{6}$$

We define the modal linewidth from the diagonal element:

$$\Gamma_n \equiv \frac{(\mathbf{V}^T\mathbf{C}\mathbf{V})_{nn}}{\mu_n} = \frac{\sum_i V_{in}^2\, m_i\, \gamma_i}{\mu_n}, \tag{7}$$

a mass-weighted average of the per-oscillator damping rates over the mode shape. For uniform damping $\gamma_i = \gamma$ this reduces to $\Gamma_n = \gamma$ for all n.

The modal equations are fully decoupled only when the off-diagonal elements of $\mathbf{V}^T\mathbf{C}\mathbf{V}$ are negligible:

$$\sum_i V_{in}\, m_i\, \gamma_i\, V_{im} \approx 0 \qquad (m \neq n). \tag{8}$$

This holds exactly for uniform γ_i and approximately when modes are well separated in frequency relative to Γ_n. When eq. (8) does not hold the full state-space treatment is required.

3.3 Modal susceptibility

Under eq. (8), the generalized force on mode n,

$$Q_n(t) = \mathbf{v}_n^T\mathbf{f}(t) = \sum_i V_{in}\, f_i(t), \tag{9}$$

drives q_n independently via

$$\tilde{q}_n(f) = \chi_n(f)\, \tilde{Q}_n(f), \tag{10}$$

where the modal susceptibility is

$$\chi_n(f) = \frac{1}{\mu_n\big[(2\pi f_n)^2 - (2\pi f)^2 + i\,(2\pi f)\,(2\pi\Gamma_n)\big]}. \tag{11}$$

The quality factor of mode n is f_n/Γ_n, note that Γ_n is in Hz and not rad/s.

3.4 Observable

A physical readout (interferometer, camera, or capacitive sensor) produces a single scalar time series

$$O(t) = \sum_i w_i\, x_i(t) = \mathbf{w}^T\mathbf{x}(t), \tag{12}$$

where $\mathbf{w}$ is the readout weight vector (discussed in Section 8). Substituting eq. (2):

$$O(t) = \sum_n N_n\, q_n(t), \tag{13}$$

where the modal readout coupling is

$$N_n \equiv \mathbf{w}^T\mathbf{v}_n = \sum_i w_i\, V_{in}. \tag{14}$$

If $N_n = 0$ for some mode, that mode is invisible to the readout regardless of how strongly it is driven.

4 SNR and strain-referred sensitivity

For a known waveform $h(t)$, the matched-filter SNR accumulated over the observation band is (Maggiore 2007):

$$\rho^2 = 4 \int_0^\infty \frac{|\tilde{O}^{\rm sig}(f)|^2}{S_O^{\rm noise}(f)}\, df = 4 \int_0^\infty \frac{|T(f)|^2\, |\tilde{h}(f)|^2}{S_O^{\rm noise}(f)}\, df, \tag{15}$$

where $S_O^{\rm noise}(f)$ is the total noise PSD at the observable. This can be written equivalently as

$$\rho^2 = 4 \int_0^\infty \frac{|\tilde{h}(f)|^2}{S_h(f)}\, df, \tag{16}$$

where the *strain-referred noise PSD* is

$$S_h(f) \equiv \frac{S_O^{\rm noise}(f)}{|T(f)|^2}. \tag{17}$$

The square root $\sqrt{S_h(f)}$ is the sensitivity curve. We now compute $T(f)$ and $S_O^{\rm noise}(f)$ in turn.

The total noise at the observable has two independent contributions,

$$S_O^{\rm noise}(f) = S_O^{\rm th}(f) + S_O^{\rm ro}(f), \tag{18}$$

where $S_O^{\rm th}$ is thermal (Brownian) force noise and $S_O^{\rm ro}$ is readout (shot) noise. These are derived in Sections 7 and 8 respectively.

5 Signal transfer function

A gravitational wave of strain $h(t)$ exerts a force on oscillator i:

$$f_i^{\rm sig}(t) = \beta_i\, h(t), \tag{19}$$

where for a particle of mass m_i with trap frequency $f_i^{\rm trap}$ in a cavity of length L:

$$\beta_i = m_i\, (2\pi f_i^{\rm trap})^2\, L. \tag{20}$$

This follows from the tidal GW force in the long-wavelength limit: in the frame of the trap centre, the particle experiences a differential displacement $\delta x_i^{\rm eq} = hL/2$, so the restoring force $m_i\omega_i^2 \delta x_i^{\rm eq}$ gives $\beta_i = m_i \omega_i^2 L$ (valid when $f_i^{\rm trap} \gg f_{\rm GW}$, i.e. the particle tracks its trap rather than free-falling).

The generalized signal force on mode n is

$$Q_n^{\rm sig}(t) = \mathcal{B}_n\, h(t), \qquad \mathcal{B}_n \equiv \mathbf{v}_n^T \boldsymbol{\beta} = \sum_i V_{in}\, \beta_i. \tag{21}$$

The signal at the observable is

$$\tilde{O}^{\text{sig}}(f) = T(f)\,\tilde{h}(f), \tag{22}$$

with transfer function

$$T(f) = \sum_{n=1}^{N} N_n\,\mathcal{B}_n\,\chi_n(f). \tag{23}$$

6 Noise

A random force on oscillator i drives all oscillators through the mechanical coupling, so one cannot write $S_O^{\text{th}} = \sum_i w_i^2\,S_i^F$ directly. Under the diagonal damping assumption eq. (8), however, each mode responds independently to its own generalized force Q_n, reducing the coupled problem to N independent scalar ones.

Let $S_{ij}^F(f)$ be the cross-spectral density of force noise between oscillators i and j. The modal force noise PSD is

$$S_{Q_n}(f) = \sum_{i,j} V_{in}\,S_{ij}^F(f)\,V_{jn}. \tag{24}$$

Under eq. (8) the modes are statistically independent, so the thermal noise PSD at the observable is an incoherent sum:

$$S_O^{\text{th}}(f) = \sum_{n=1}^{N} N_n^2\,|\chi_n(f)|^2\,S_{Q_n}(f). \tag{25}$$

7 Uncorrelated forces in UHV

For levitated particles in UHV, residual-gas collisions are spatially local and uncorrelated between particles: $S_{ij}^F = S_i^F\,\delta_{ij}$. Applying the fluctuation-dissipation theorem per oscillator gives

$$S_i^F = 4\,k_B T\,m_i\,\gamma_i. \tag{26}$$

Substituting into eq. (24):

$$S_{Q_n}^{\text{th}} = 4k_B T \sum_i V_{in}^2\,m_i\,\gamma_i = 4k_B T\,\mu_n\,\Gamma_n, \tag{27}$$

where the second equality uses eq. (7). The thermal noise at the observable is therefore

$$S_O^{\text{th}}(f) = 4k_B T \sum_{n=1}^{N} N_n^2\,|\chi_n(f)|^2\,\mu_n\,\Gamma_n. \tag{28}$$

The corresponding strain-referred thermal noise is

$$\boxed{S_h^{\mathrm{th}}(f) = \frac{4k_BT\displaystyle\sum_{n=1}^{N} N_n^2\,|\chi_n(f)|^2\,\mu_n\,\Gamma_n}{\left|\displaystyle\sum_n N_n\,\mathcal{B}_n\,\chi_n(f)\right|^2}.} \tag{29}$$

In the limit where a single mode n dominates near its resonance, the N_n^2 cancels between numerator and denominator and S_h^{th} becomes independent of the readout direction $\mathbf{w}$.

8 Readout noise

We consider two readout strategies for an array of particles along $\hat{x}$.

8.1 Readout A: cavity probe

The cavity probe wavelength $\lambda_{\mathrm{probe}} = 1064\,\mathrm{nm}$ (angular frequency $\omega_L = 2\pi c/\lambda_{\mathrm{probe}}$) propagates along $\hat{x}$ and accumulates phase from each particle's displacement. From input-output theory for a resonantly driven, impedance-matched cavity with decay rate κ, the phase quadrature of the reflected field at sideband frequency f is

$$\tilde{\phi}(f) = \mathcal{F}(f)\sum_i g_i\,\tilde{x}_i(f), \qquad \mathcal{F}(f) = \frac{2/\kappa}{1 + if/f_{\mathrm{cav}}}, \qquad f_{\mathrm{cav}} = \frac{\kappa}{2\pi}, \tag{30}$$

where $g_i = \partial\omega_{\mathrm{cav}}/\partial x_i$ $[\mathrm{rad\,s^{-1}\,m^{-1}}]$ is the optomechanical coupling at site i and $\mathcal{F}(f)$ is the cavity low-pass filter. The transduction is flat for $f \ll f_{\mathrm{cav}}$ and rolls off above the cavity pole.

We define the displacement-referred observable by dividing out the cavity response:

$$\widetilde{O}(f) \equiv \frac{\tilde{\phi}(f)}{\mathcal{F}(f)\cdot|\boldsymbol{g}|}, \qquad |\boldsymbol{g}| \equiv \sqrt{\sum_i g_i^2}. \tag{31}$$

Since $\mathcal{F}(f)$ cancels algebraically between numerator and denominator, this reduces to

$$O(t) = \sum_i w_i\,x_i(t), \qquad w_i = \frac{g_i}{|\boldsymbol{g}|}, \qquad \|\mathbf{w}\| = 1. \tag{32}$$

The readout weights w_i are fixed by the intracavity field profile at each particle's equilibrium position and are not freely tunable.

The shot noise on the phase measurement is white:

$$S_\phi = \frac{\hbar\omega_L}{\eta_{\mathrm{det}}\,P_{\mathrm{det}}} \quad [\mathrm{rad^2\,Hz^{-1}}], \tag{33}$$

where P_{det} is the detected power and η_{det} the detector quantum efficiency. Shot noise enters at the photodetector, after the cavity has acted. Referring it back to the observable O via eq. (31):

$$S_O^{\text{ro,A}}(f) = \frac{S_\phi}{|\mathcal{F}(f)|^2 \cdot |\boldsymbol{g}|^2} = \frac{S_\phi}{|\boldsymbol{g}|^2}\left(\frac{\kappa}{2}\right)^2 \left[1 + \left(\frac{f}{f_{\text{cav}}}\right)^2\right]. \tag{34}$$

This rises above f_{cav} because the cavity provides less phase transduction at high frequencies: a fixed shot noise floor S_ϕ corresponds to a larger inferred displacement uncertainty when the cavity response is weaker.

The strain-referred shot noise contribution is:

$$\boxed{S_h^{\text{ro,A}}(f) = \frac{\hbar\omega_L}{\eta_{\text{det}}\, P_{\text{det}}} \cdot \frac{1}{\sum_i g_i^2} \cdot \frac{\frac{\kappa^2}{4}\left[1 + \left(\frac{f}{f_{\text{cav}}}\right)^2\right]}{\left|\sum_n N_n\, \mathcal{B}_n\, \chi_n(f)\right|^2},} \tag{35}$$

where $f_{\text{cav}} = \kappa/(2\pi)$ is the cavity pole frequency and $N_n = \sum_i (g_i/|\boldsymbol{g}|)\, V_{in}$ are the modal readout couplings from eq. (14) with weights $w_i = g_i/|\boldsymbol{g}|$. Both the readout coupling N_n and the signal coupling $\mathcal{B}_n$ must be nonzero for mode n to contribute.

Since the mechanical signal, the thermal noise, and the readout weights all carry the same $\mathcal{F}(f)$ factor, it cancels in both $T(f)$ and S_O^{th}, which retain the standard modal-sum forms eqs. (23) and (28). The cavity filter appears only in $S_O^{\text{ro,A}}$, and hence the shot-noise-limited strain sensitivity degrades above f_{cav} as $1 + (f/f_{\text{cav}})^2$, while the thermal-noise-limited sensitivity is unaffected by the cavity bandwidth.

8.2 Readout B: transverse imaging

A probe beam at wavelength $\lambda_{\text{ro}} = 532\,\text{nm}$ propagates along $\hat{z}$ (perpendicular to the array). With inter-particle spacing $d = 10\lambda_{\text{trap}} = 15.5\,\mu\text{m}$ and $\lambda_{\text{ro}} \ll d$, a modest numerical aperture of $\text{NA} = 0.1$ gives a diffraction limit of $\lambda_{\text{ro}}/(2\,\text{NA}) \approx 2.7\,\mu\text{m} \ll d$, so each particle is resolved independently. The N particle images fall on separate camera regions, giving N independent displacement measurements with freely configurable weights w_i.

The shot-noise-limited displacement sensitivity per particle follows from the deflection of a focused Gaussian beam of waist $w_0 = \lambda_{\text{ro}}/(\pi\,\text{NA})$ by a split detector. A displacement δx_i shifts the beam at the detector; differentiating the power on one half with respect to δx_i and evaluating the resulting Gaussian integral gives the transduction gradient

$$\left.\frac{d(P_R - P_L)}{d(\delta x_i)}\right|_0 = \frac{2\sqrt{2}}{\sqrt{\pi}}\,\frac{P_i^{\text{sc}}}{w_0}. \tag{36}$$

The prefactor follows from $P_R - P_L = P_i^{\rm sc}\,{\rm erf}(x/(\sqrt{2}w_0))$, whose derivative at $x = 0$ is $P_i^{\rm sc} \cdot (2/\sqrt{\pi}) \cdot (1/(\sqrt{2}w_0)) \cdot \sqrt{2} = (2\sqrt{2}/\sqrt{\pi})\, P_i^{\rm sc}/w_0$.

The one-sided shot-noise PSD on the difference photocurrent, referred to equivalent power, is $S_{\Delta P} = 2\hbar\omega_{\rm ro}\, P_i^{\rm sc}/\eta_{\rm det}$. Dividing by the squared transduction gradient:

$$S_i^{\rm ro} = \frac{S_{\Delta P}}{\big(d(P_R - P_L)/d(\delta x_i)\big)^2} = \frac{\pi\, \hbar\omega_{\rm ro}\, w_0^2}{4\, \eta_{\rm det}\, P_i^{\rm sc}}. \tag{37}$$

Substituting $w_0 = \lambda_{\rm ro}/(\pi\, {\rm NA})$:

$$S_i^{\rm ro} = \frac{\hbar\omega_{\rm ro}\, \lambda_{\rm ro}^2}{4\pi\, \eta_{\rm det}\, {\rm NA}^2\, P_i^{\rm sc}}, \tag{38}$$

where $P_i^{\rm sc}$ is the power collected from particle i. The readout noise at the observable is

$$S_O^{\rm ro,B} = \sum_i w_i^2\, S_i^{\rm ro}. \tag{39}$$

The weights w_i and noise levels $S_i^{\rm ro}$ are independent: w_i is a free post-processing choice, while $S_i^{\rm ro}$ depends only on the imaging hardware. If the probe power is allocated as $P_i^{\rm sc} \propto w_i^2$, then $S_i^{\rm ro} \propto 1/w_i^2$ and $S_O^{\rm ro,B}$ is independent of the weight distribution.

Dividing by $|T(f)|^2$ from eqs. (17) and (23) gives the strain-referred readout noise for Readout B:

$$\boxed{S_h^{\rm ro,B}(f) = \frac{\displaystyle\sum_i w_i^2\, S_i^{\rm ro}}{\left|\displaystyle\sum_n N_n\, \mathcal{B}_n\, \chi_n(f)\right|^2}, \qquad S_i^{\rm ro} = \frac{\hbar\omega_{\rm ro}\, \lambda_{\rm ro}^2}{4\pi\, \eta_{\rm det}\, {\rm NA}^2\, P_i^{\rm sc}}.} \tag{40}$$

In the narrow-linewidth limit ($\Gamma_n \ll f_n$, well-separated modes), evaluating at $f = f_n$ using $|\chi_n(f_n)|^2$ from eq. (46):

$$S_h^{\rm ro,B}(f_n) = \frac{(2\pi)^4 f_n^2\, \Gamma_n^2\, \mu_n^2 \displaystyle\sum_i w_i^2\, S_i^{\rm ro}}{N_n^2\, \mathcal{B}_n^2}. \tag{41}$$

The shot-noise contribution to S_h scales as Γ_n^2: high-Q modes benefit from their enhanced mechanical response.

8.3 Comparison

	Readout A	Readout B
Geometry	Axial cavity probe	Transverse imaging
Probe wavelength	$\lambda_{\text{probe}} = 1064\,\text{nm}$	$\lambda_{\text{ro}} = 532\,\text{nm}$
w_i set by	g_i (fixed by physics)	Post-processing (free)
Independent channels	1	N
w_i and S_i^{ro} independent?	No (single channel)	Yes

9 Inspiral signal: Maggiore's formula

For a compact binary inspiral, the Fourier transforms of the two polarisations are given by the stationary-phase approximation (Maggiore 2007, Ch. 4):

$$\tilde{h}_+(f) = \frac{1}{\pi^{2/3}}\sqrt{\frac{5}{24}}\,\frac{c}{r}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/6}\frac{1+\cos^2\iota}{2}\,\frac{e^{i\Psi_+(f)}}{f^{7/6}}, \tag{42}$$

$$\tilde{h}_\times(f) = \frac{1}{\pi^{2/3}}\sqrt{\frac{5}{24}}\,\frac{c}{r}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/6}\cos\iota\,\frac{e^{i\Psi_\times(f)}}{f^{7/6}}, \tag{43}$$

where $\mathcal{M}_c$ is the chirp mass, r the luminosity distance, ι the inclination angle, and the phases satisfy $\Psi_\times = \Psi_+ + \pi/2$. Taking modulus squared and summing polarisations (or treating a single polarisation for a linearly polarised source), the key result is that

$$|\tilde{h}(f)|^2 = \mathcal{A}^2\, f^{-7/3}, \tag{44}$$

where $\mathcal{A}^2$ collects all source-dependent constants:

$$\mathcal{A}^2 \equiv \frac{5}{24\pi^{4/3}}\,\frac{c^2}{r^2}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/3}\Theta(\iota). \tag{45}$$

Here $\Theta(\iota)$ is a dimensionless inclination factor; for optimal orientation $\Theta = 1$.

Slowly varying approximation. The formula eq. (44) assumes the amplitude $\mathcal{A}$ is approximately constant over a resonance width Γ_n. Since $\mathcal{A}^2 f^{-7/3}$ varies on the scale of f_n, the condition is simply $\Gamma_n \ll (3/7)f_n$, which is automatically satisfied for any high-Q mode and is already subsumed by $\Gamma_n \ll f_n$.

10 Narrow-linewidth limit

When $\Gamma_n \ll f_n$ and mode frequencies are well separated ($|f_m - f_n| \gg \Gamma_n$ for $m \neq n$), the Lorentzian integrals evaluate (by residues) to

$$\int_0^\infty |\chi_n(f)|^2 \, df = \frac{1}{4\,\mu_n^2\,(2\pi)^3\,f_n^2\,\Gamma_n}, \qquad |\chi_n(f_n)|^2 = \frac{1}{\mu_n^2\,(2\pi)^4\,f_n^2\,\Gamma_n^2}. \tag{46}$$

The per-mode contributions to S_h evaluated at $f = f_n$ are collected in Table 1. For thermal noise the N_n^2 cancels between $S_O^{\rm th}$ and $|T|^2$, leaving $S_h^{\rm th}(f_n) = 4k_B T \mu_n \Gamma_n / \mathcal{B}_n^2$, which is flat near f_n with no Lorentzian structure. For the readout-noise case, $S_h^{\rm ro}$ has a Lorentzian dip at each f_n, so the integrand $|\tilde{h}|^2/S_h^{\rm ro}$ is peaked there with width Γ_n. The per-mode SNR decomposition $\rho^2 = \sum_n \rho_n^2$ therefore applies cleanly to the *readout-noise-limited* case: the integral over each peak converges independently and is evaluated using eq. (46).

For the *thermal-noise-limited* case the situation is different. $S_h^{\rm th}$ is flat near each f_n (the Lorentzian in $|\chi_n|^2$ cancels between numerator and denominator), so the integrand $|\tilde{h}|^2/S_h^{\rm th} \propto f^{-7/3}$ has no peak at any mode frequency. The SNR integral does not localise near individual modes and a per-mode decomposition is not meaningful. The thermal-noise-limited SNR must be evaluated as a single integral over the full band:

$$\rho^2\big|_{\rm th} = 4\int_0^\infty \frac{\mathcal{A}^2\, f^{-7/3}}{S_h^{\rm th}(f)}\, df, \tag{47}$$

with $S_h^{\rm th}(f)$ from eq. (29).

10.1 Per-mode SNR and weight optimisation

In the readout-noise-limited case, the matched-filter SNR accumulates independently from each mode:

$$\rho^2\big|_{\rm ro} = \sum_n \rho_n^2, \tag{48}$$

where each ρ_n^2 is given in Table 2.

Validity conditions. The decomposition eq. (48) requires $\Gamma_n \ll f_n$ (so that the Lorentzian integral is well converged) and well-separated mode frequencies ($|f_m - f_n| \gg \Gamma_n$ for all $m \neq n$, so that cross-mode interference terms in $|T(f)|^2$ are suppressed at each resonance).

The double sum in Readout B. For Readout B the shot noise $S_O^{\rm ro,B} = \sum_i w_i^2 S_i^{\rm ro}$ is diagonal in particle space, while the mechanical response is diagonal in mode space. Since n-space and i-space are related by $\mathbf{V}$, these two diagonalisations are generically incompatible, and the per-mode SNR contains both a sum over particles in the numerator ($N_n = \sum_i w_i V_{in}$) and a sum over particles in the denominator ($\sum_i w_i^2 S_i^{\rm ro}$), nested

Table 1: Strain-referred noise PSD $S_h(f)$ from each source. *General*: valid at any f with $T(f) = \sum_n N_n \mathcal{B}_n \chi_n(f)$ from eq. (23). *Narrow-linewidth*: evaluated at $f = f_n$, assuming $\Gamma_n \ll f_n$ and well-separated modes; the factor $[1 + (f_n/f_{\rm cav})^2] \to 1$ when $f_n \ll f_{\rm cav}$.

Noise source	General	Narrow-linewidth ($f = f_n$)
Thermal	$\dfrac{4k_B T \sum_n N_n^2 \lvert\chi_n\rvert^2 \mu_n \Gamma_n}{\left\lvert \sum_n N_n \mathcal{B}_n \chi_n \right\rvert^2}$	$\dfrac{4k_B T \mu_n \Gamma_n}{\mathcal{B}_n^2}$
Readout A (cavity)	$\dfrac{\hbar\omega_L\,(\kappa/2)^2 \left[1 + \left(\dfrac{f}{f_{\rm cav}}\right)^2\right]}{\eta_{\rm det}\, P_{\rm det}\, \lvert\boldsymbol{g}\rvert^2 \left\lvert \sum_n N_n \mathcal{B}_n \chi_n \right\rvert^2}$	$\dfrac{\hbar\omega_L\,\kappa^2\,\mu_n^2 (2\pi)^4 f_n^2 \Gamma_n^2}{4\,\eta_{\rm det}\, P_{\rm det}\, \lvert\boldsymbol{g}\rvert^2\, N_n^2\, \mathcal{B}_n^2} \left[1 + \left(\dfrac{f_n}{f_{\rm cav}}\right)^2\right]$
Readout B (imaging)	$\dfrac{\sum_i w_i^2 \dfrac{\hbar\omega_{\rm ro}\lambda_{\rm ro}^2}{4\pi\,\eta_{\rm det}\,{\rm NA}^2\, P_i^{\rm sc}}}{\left\lvert \sum_n N_n \mathcal{B}_n \chi_n \right\rvert^2}$	$\dfrac{(2\pi)^4 f_n^2 \Gamma_n^2 \mu_n^2}{N_n^2\, \mathcal{B}_n^2} \sum_i \dfrac{w_i^2\, \hbar\omega_{\rm ro}\lambda_{\rm ro}^2}{4\pi\,\eta_{\rm det}\,{\rm NA}^2\, P_i^{\rm sc}}$

inside the outer sum over modes. The double sum is therefore a consequence of the basis mismatch between readout and mechanics. It collapses to a single sum when the per-particle noise is uniform ($S_i^{\rm ro} = S^{\rm ro}$ for all i), in which case $\sum_i w_i^2 S_i^{\rm ro} = S^{\rm ro}\|\mathbf{w}\|^2$ is a scalar and mode- and particle-sums decouple.

Optimal weights and the Cauchy–Schwarz bound. For a given target mode n the ratio $N_n^2/\sum_i w_i^2 S_i^{\rm ro}$ in $\rho_n^2|_{\rm ro,B}$ is maximised over the weight vector $\mathbf{w}$ by the Cauchy–Schwarz inequality. Writing $N_n = \sum_i w_i V_{in}$ and applying Cauchy–Schwarz with the inner product weighted by $S_i^{\rm ro}$:

$$\frac{N_n^2}{\sum_i w_i^2\, S_i^{\rm ro}} \;\le\; \sum_i \frac{V_{in}^2}{S_i^{\rm ro}}, \tag{49}$$

with equality when $w_i^\star \propto V_{in}/S_i^{\rm ro}$. At the optimum the double sum collapses entirely:

$$\rho_n^{2,\star}\big|_{\rm ro,B} = \frac{\mathcal{A}^2\, f_n^{-7/3}\, \mathcal{B}_n^2}{\mu_n^2\, (2\pi)^3\, f_n^2\, \Gamma_n} \sum_i \frac{V_{in}^2}{S_i^{\rm ro}}. \tag{50}$$

The structure is now a single sum over particles, directly analogous to Readout A where $\sum_i g_i^2$ (the collective cavity coupling) plays the role of $\sum_i V_{in}^2/S_i^{\mathrm{ro}}$. In both cases the effective figure of merit is a particle sum weighted by how strongly each particle participates in mode n (V_{in}^2), divided by the per-particle noise cost.

The optimal weights $w_i^\star \propto V_{in}/S_i^{\mathrm{ro}}$ are mode-specific: they depend on n through the mode shape V_{in}. Consequently, the weights that maximise ρ_n^2 for one mode will generically degrade ρ_m^2 for $m \neq n$. Simultaneous optimisation over all modes is a quadratic programme in $\mathbf{w}$, and the solution trades off sensitivity across modes. This mode-specificity is invisible in the $S_h(f)$ table but becomes explicit in the ρ^2 table, and is a key practical consideration when the signal is expected near a particular mode.

Table 2: Per-mode SNR ρ_n^2 in the readout-limited, narrow-linewidth limit ($\Gamma_n \ll f_n$, well-separated modes). The signal amplitude uses Maggiore's inspiral formula $|\tilde{h}(f_n)|^2 = \mathcal{A}^2 f_n^{-7/3}$ (eq. (44)). The total readout-limited SNR is $\rho^2 = \sum_n \rho_n^2$ (eq. (48)); for the thermal-noise-limited case use the full integral eq. (47). The expressions follow from the Lorentzian integral eq. (46). For Readout B the optimal-weight row gives the Cauchy–Schwarz upper bound eqs. (49) and (50), attained at $w_i^\star \propto V_{in}/S_i^{\mathrm{ro}}$.

Noise source	ρ_n^2
Readout A (cavity)	$\dfrac{\eta_{\mathrm{det}}\, P_{\mathrm{det}} \left(\sum_i g_i V_{in}\right)^2 \mathcal{B}_n^2\, \mathcal{A}^2\, f_n^{-7/3}}{\mu_n^2\, (2\pi)^3\, f_n^2\, \Gamma_n\, \hbar\omega_L\, (\kappa/2)^2}$
Readout B, general w_i	$\dfrac{N_n^2\, \mathcal{B}_n^2\, \mathcal{A}^2\, f_n^{-7/3}}{\mu_n^2\, (2\pi)^3\, f_n^2\, \Gamma_n \sum_i w_i^2\, \dfrac{\hbar\omega_{\mathrm{ro}}\lambda_{\mathrm{ro}}^2}{4\pi\, \eta_{\mathrm{det}}\, \mathrm{NA}^2\, P_i^{\mathrm{sc}}}}$ $\quad (N_n = \sum_i w_i V_{in})$
Readout B, optimal $w_i^\star \propto V_{in}/S_i^{\mathrm{ro}}$	$\dfrac{\mathcal{B}_n^2\, \mathcal{A}^2\, f_n^{-7/3}}{\mu_n^2\, (2\pi)^3\, f_n^2\, \Gamma_n} \sum_i \dfrac{4\pi\, \eta_{\mathrm{det}}\, \mathrm{NA}^2\, P_i^{\mathrm{sc}}\, V_{in}^2}{\hbar\omega_{\mathrm{ro}}\, \lambda_{\mathrm{ro}}^2}$